(باللغة العربية):

Section or Time:

Question 1: Circle the most right answer that describes your answer [50, Points]

_1) Which of the following statements is false?
a) a geometric series is a p-series
b) a geometric series can be convergent
c) every power series is convergent at its center
d) every Taylor series is a power series
e) none

_2) Which of the following statements is true for $\frac{1}{2}-\frac{1}{4}+\frac{1}{8}-\frac{1}{16}+\frac{1}{32}-\frac{1}{64}+\frac{1}{128}+\cdots$:
a) absolutely convergent
b) conditionally converge
c) neither absolutely nor conditionally convergent
d) divergent
e) none

_3) If $\sum_{k=1}^{\infty} 8a_k = 15$, then the value of $\lim_{k\to\infty}(4a_k - 1)$ equals:
a) $\frac{15}{8}$
b) 3
c) ∞
d) -1
e) none

_4) Determine which of the following statements applies to $\sum_{k=1}^{\infty}\frac{(-3)^{2k}}{k^k}$:
a) diverges by the root test
b) converges by the alternating series test
c) diverges by the alternating series test
d) converges by the root test
e) none

_5) Which of the following expressions is true for $\left\{\frac{1}{\sqrt[5]{n^2+1}}\right\}_{n=1}^{\infty}$:
a) decreasing and converge
b) increasing and diverge
c) increasing and converge
d) decreasing and diverge
e) none

_6) The series $\sum_{k=1}^{\infty} k^{\frac{1}{k}}$ is:
a) converges
b) diverges
c) converges absolutely
d) converges conditionally
e) none

_7) For $\sum_{k=1}^{\infty}\frac{1}{1+\left(\frac{1}{2}\right)^k}$ one of the following statements is true:
a) convergent by comparison with $\sum_{k=1}^{\infty}\left(\frac{1}{2}\right)^k$
b) convergent because it is a geometric series and $r=\frac{1}{2}$
c) diverge by the divergent test
d) divergent by the integral test
e) none

_8) The value of $\lim_{n\to\infty}\left(\frac{n+3}{n}\right)^n$ equals:
a) e^{-3}
b) $e^{\left(\frac{1}{3}\right)}$
c) $e^{-\left(\frac{1}{3}\right)}$
d) e^3
e) none

_) The Taylor series of $\sin(x)$ at $x=\frac{\pi}{2}$ is:
a) $\sum_{k=0}^{\infty}(-1)^k\frac{\left(x-\frac{\pi}{2}\right)^{2k+1}}{(2k+1)!}$
b) $\sum_{k=0}^{\infty}\frac{\left(x-\frac{\pi}{2}\right)^{2k+1}}{(2k+1)!}$
c) $\sum_{k=0}^{\infty}\frac{\left(x-\frac{\pi}{2}\right)^{2k}}{(2k)!}$
d) $\sum_{k=0}^{\infty}(-1)^k\frac{\left(x-\frac{\pi}{2}\right)^{2k}}{(2k)!}$
e) none

_10) If $\sum_{k=0}^{\infty} r^k = 3$, the the value of r will be:

a) $-\frac{3}{2}$ b) $\frac{2}{3}$ c) $-\frac{2}{3}$ d) $\frac{3}{2}$ e) none

_11) The interval of convergence for $\sum_{k=0}^{\infty}(-1)^{k+1}\frac{(x+1)^k}{k+1}$ is:

a) $[-2,0]$ b) $(-2,0]$ c) $[-2,0)$ d) $(-2,0)$ e) none

For Questions 12, 13, and [illegible]:

Given the Taylor series of e^x about $x = 0$ as $e^x = \sum_{k=0}^{\infty}\frac{x^k}{k!}, -\infty < x < \infty$:

_12) The Maclaurin series of xe^x has the form:

a) $\sum_{k=0}^{\infty}\frac{x^{k-1}}{k!}$ b) $\sum_{k=1}^{\infty}\frac{x^{k+1}}{k!}$ c) $\sum_{k=0}^{\infty}\frac{x^{k+1}}{k!}$ d) $\sum_{k=1}^{\infty}\frac{x^{k-1}}{k!}$ e) none

_13) The sum of the series $\sum_{k=0}^{\infty}\frac{k+1}{k!}$ equals:

a) e b) $\frac{2}{e}$ c) $2e$ d) $\frac{1}{e}$ e) none

_14) The approximate value of $\int_0^1 e^{x^2}dx$ has the form:

a) $\sum_{k=0}^{\infty}\frac{1}{(2k+1)k!}$ b) $\sum_{k=0}^{\infty}\frac{1}{(2k-1)k!}$ c) $\sum_{k=0}^{\infty}\frac{(-1)^k}{(2k+1)k!}$ d) $\sum_{k=0}^{\infty}\frac{(-1)^k}{(2k-1)k!}$ e) none

_15) The polar coordinates of $(x,y) = (-2,2\sqrt{3})$ is:

a) $\left(4,\frac{\pi}{6}\right)$ b) $\left(4,\frac{2\pi}{3}\right)$ c) $\left(-4,\frac{2\pi}{3}\right)$ d) $\left(-4,\frac{\pi}{6}\right)$ e) none

_16) The form of $x^2 + y^2 - 6y = 0$ in polar coordinates is:

a) $r = 3\sin\theta$ b) $r = 3\cos\theta$ c) $r = 6\sin\theta$ d) $r = 6\cos\theta$ e) none

_17) The value of $\int_e^{\infty}\frac{(\ln(x))^{-4}}{x}dx$ is:

a) converges to $-\frac{1}{3}$ b) converges to 0 c) diverges [illegible] converges to $\frac{1}{3}$ e) none

_18) The form of $\int \sin(x)\cos\left(\frac{1}{2}x\right)dx$ is:

a) $\frac{1}{5}\cos(5x) + \frac{1}{2}c + c$ b) $\cos(5x) - \frac{1}{2}\cos(x) + c$

c) $-\frac{1}{3}\cos\left(\frac{3}{2}x\right) + \cos\left(\frac{1}{2}x\right) + c$ d) $-\frac{1}{3}\cos\left(\frac{3}{2}x\right) - \cos\left(\frac{1}{2}x\right) + c$ e) none

_19) The value of sech(ln 2) equals:

a) $-\frac{4}{3}$ b) $-\frac{4}{5}$ c) $\frac{4}{5}$ d) $\frac{4}{3}$ e) none

_20) The setup of the integral that gives the area of the region enclosed by $x = y^2$ and $y = x - 2$, integrating with respect to y will be:

a) $2\int_0^1 \sqrt{x}\,dx + \int_1^4(\sqrt{x} - x + 2)\,dx$ b) $\int_{-1}^{2}(y + 2 - y^2)\,dy$

c) $\int_{-1}^{2}(y + 2 + y^2)\,dy$ d) $2\int_0^1 \sqrt{x}\,dx - \int_1^4(\sqrt{x} - x + 2)\,dx$ e) none

_21) The form of $\frac{d}{dy}\cosh(y)$ is:

a) $\sinh(y)$ b) $\sin(y)$ c) $\cosh(y)$ d) $\cos(y)$ e) none